

DBMS Mid-Sem Question Paper 2025

C.V. Raman Global University • Computer Science & Engineering • Semester 3

SECTION A: Conceptual & Relational Queries [15 Marks]

1. What is the fundamental difference between relational algebra and relational calculus? Write the relational algebra expression to find names of all students enrolled in Database Systems.
2. Differentiate between Theta Join and Natural Join with a schema example.
3. Explain the concept of total participation and partial participation in an Entity-Relationship (ER) model with a real-world diagram.

SECTION B: Advanced SQL & Schema Design [15 Marks]

4. Consider Schema: Employee(emp_id, emp_name, dept_id, salary) and Department(dept_id, dept_name, location).

Write SQL query to find the second highest salary in each department using Window functions.

5. Define ON DELETE CASCADE and describe scenarios where it must be avoided to prevent accidental cascading data loss.

SECTION C: Functional Dependencies & Normalization [20 Marks]

6. Consider relation R(A, B, C, D, E) with F = { A → BC, CD → E, B → D, E → A }.

- (a) Identify all candidate keys of relation R.
- (b) Determine the highest normal form of R (1NF, 2NF, 3NF, or BCNF).
- (c) Decompose R into BCNF ensuring lossless join property.